

THE RELATIONAL MODEL, SUCCEEDED FOR DERIVED KNOWLEDGE

A Successor to Codd

A formal result, proven in code. Edgar Codd, 1970, to the AI era.

In 1970, Edgar Codd defined the relational model: data as relations, accessed declaratively, decomposed without loss. It became the foundation of nearly every system of record built since. Its primitive, the tuple, was deliberately provenance free. A fact carried no memory of where it came from. In 1970 that was the correct choice. It bought data independence for a single trusted data bank.

In a world where knowledge is derived by machines, across parties that do not trust each other, a provenance free fact is the liability, not the feature. The anchor and submodule model inverts Codd's primitive on exactly this axis. It is the relational model's successor for derived knowledge. This document states the claim precisely and gives the proof.

What a successor must match

Codd 1970 is not tables. It is four things, and a successor must answer all four: (1) one uniform data structure; (2) a closed algebra with equivalence laws, which is what made the query optimizer possible; (3) a decomposition theory that says which decompositions are lossless; and (4) data independence, the abolition of the access path, where you state what you want and the system chooses how to reach it.

The inversion

The anchor and submodule model changes not the shape of the data but what a fact is. The served state is defined as a signed fold over an ordered provenance log: $\Sigma = \text{fold}(\text{gate}, \Lambda)$. The state is a function of its lineage, and it cannot change without a signed log event. A fact, the submodule, is a rate over that lineage: $\text{variance} = \partial \text{Base} / \partial \text{Provenance}$, the novel signal a fact introduces over the distinct provenance sources that produced it. Integrity is carried in the datum itself: every layer is doubly signed and verifies offline, with the derivative bound to the live state. Codd assumed a trusted engine around the data. This assumes none.

Every prior post relational system, document and graph and key value and lakehouse, changed the shape of the data and kept the fact provenance free. This is the first that moves provenance into the primitive itself.

The proof

We held the model against Codd's four requirements and were honest about where it fell short. On the axis Codd excluded it was already ahead. On Codd's own ground it was behind on three of four. We closed each gap in code, with passing tests.

Codd requirement	Verdict	How it is met
One uniform structure	HOLDS	Anchor to submodule, one structure used everywhere.
Set semantics, order independence	BY DESIGN	The state is an ordered signed fold, so temporality is first class. Proven commutative when no gate binds.
Data independence, no access path	EARNED	A predicate select over a row relation. A fact is reachable by what it is, with its anchor returned as output. Path independent.

Closed algebra with equivalence laws	EARNED	Join, union, project, select, closed over the relation, with proven laws: selection commutativity, projection, union and join, variance composition.
Lossless decomposition	EARNED	Reform routes every unmatched record to a residual, so reform is a total partition. Nothing is dropped.
Integrity	EXCEEDS	Doubly signed, offline verifiable, the derivative bound to live state.
Provenance	EXCEEDS	The state is a signed fold over the provenance log. A fact is a rate over lineage. Beyond Codd and beyond the provenance literature.

The three theorems, proven in code

A. Declarative access. A submodule is reached by a predicate over its fields, with its anchor returned as an output rather than required as input. The result is path independent: equal to an independent traversal and invariant under order. This abolishes the access path that the model previously relied on.

B. A closed algebra with laws. The binary operators the model lacked, join and union, together with projection and selection, closed over the relation, with proven equivalence laws. The central law: variance composition is the provenance set union law, the homomorphic image of how provenance. This is the precise bridge to provenance semirings (Green, Karvounarakis and Tannen, 2007). That work annotates Codd's relations so provenance composes through the algebra. Here the composing quotient is the primitive itself.

C. Lossless reform. Restructuring routes every record that matches no target to a residual module, making reform a total partition: the union of all module members equals the input set. The relational model is famous for lossless decomposition. This is its analog, and it holds.

Full system suite: **554 tests passing, zero regressions.** The three theorems contribute 20 tests, all green. On Codd's own ground the model moved from two requirements missing and one partial to three earned, each backed by a passing test. The provenance as denominator core, which already went beyond the provenance semiring literature, now stands on a declarative, law governed, lossless base.

The claim, stated precisely

This is not a replacement for the relational model where the relational model is optimal, which is ground truth facts in a trusted store. It is the relational model's successor for derived knowledge, the regime that now dominates: knowledge synthesized by machines, across parties that do not trust each other, where lineage is the load bearing fact. The claim is narrow on purpose, and it is the stronger for it. It holds.

The full formal specification, the implementation plan, and the test suite accompany this brief. Every clause above has a passing test behind it.

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